

0478/12 – JUNE 2019

Question	Answer	Marks
6(c)	– Several/multiple bits are transmitted at a time/simultaneously – Several/multiple wires are used – Data is transmitted in both directions ... – ... at the same time/simultaneously	4

Question	Answer	Marks																											
9	1 mark per each correct parity bit: Parity bit <table><tr><td>Register A</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr></table> <table><tr><td>Register B</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table> <table><tr><td>Register C</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>	Register A	1	0	1	0	0	1	0	1	Register B	1	1	0	0	0	0	0	1	Register C	1	1	0	0	0	0	1	1	3
Register A	1	0	1	0	0	1	0	1																					
Register B	1	1	0	0	0	0	0	1																					
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3(b)	Four from: – An encryption algorithm is used – ... to scramble data – The original data is called the plain text – A key is used to encrypt the data – The key is applied to the plain text – Plain text is encrypted into cypher text	4
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Question	Answer	Marks
5	– B Three from: – Added up the number of 1's / 0's in each register – With the parity bit, two registers have an odd number of 1's / 0's – One register has an even number of 1's / 0's – Odd parity must be the parity used	4

0478/11 – NOVEMBER 2019

Question	Answer	Marks
5(a)	- α Data is sent down a single wire ... ∞ ... one bit at a time ∞ Data is sent in both directions ... ∞ ... but only one direction at a time	4
5(b)	One mark for correct byte (Byte) 2 // 01010100 Three from: ∞ Added up / counted the 1s / 0s ∞ Even parity used // 3 bytes are even ∞ Byte 2 uses odd parity // 1 byte is odd	4

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Question	Answer	Marks																		
9(a)	One mark per each correct tick <table><thead><tr><th>Statement</th><th>True (✓)</th><th>False (✓)</th></tr></thead><tbody><tr><td>Duplex data transmission can be either serial or parallel</td><td>✓</td><td></td></tr><tr><td>Duplex data transmission is when data is transmitted both ways, but only one way at a time</td><td></td><td>✓</td></tr><tr><td>Duplex data transmission is always used to connect a device to a computer</td><td></td><td>✓</td></tr><tr><td>Duplex data transmission is when data is transmitted both ways at the same time</td><td>✓</td><td></td></tr><tr><td>Duplex data transmission automatically detects any errors in data</td><td></td><td>✓</td></tr></tbody></table>	Statement	True (✓)	False (✓)	Duplex data transmission can be either serial or parallel	✓		Duplex data transmission is when data is transmitted both ways, but only one way at a time		✓	Duplex data transmission is always used to connect a device to a computer		✓	Duplex data transmission is when data is transmitted both ways at the same time	✓		Duplex data transmission automatically detects any errors in data		✓	5
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Duplex data transmission automatically detects any errors in data		✓																		
9(b)	∞ Parallel data transmission	1																		

Question	Answer	Marks
9(c)	Four from (maximum two marks per benefit): ∞ It is a universal standard ... ∞ ... so it is likely to be compatible with the computer ∞ It can only be inserted one way ... ∞ ... so there is less chance of connecting a device incorrectly ∞ It is a high-speed connection ... ∞ ... so data will be transmitted quicker ∞ It uses serial transmission ... ∞ ... so it is cheaper to manufacture/buy ∞ ... less chance of skewing / errors ∞ It doesn't require a (wireless) network ... ∞ ... therefore, can be used if a network is down ∞ It is backwards compatible ... ∞ ... so no additional technology is needed ∞ It can power the device ... ∞ ... therefore no separate source of power is needed ∞ Drivers are automatically downloaded // device is automatically identified ... ∞ ... so no need to find them online / install them manually	4

Question	Answer	Marks
10(a)	Four from: ∞ Validation method ∞ Used to check data entry ∞ Digit is calculated from data // by example ∞ Digit is appended / added to data ∞ Digit is recalculated when data has been input ∞ Digits are compared ∞ If digits are different, error is detected // If digits match, no error is detected	4

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1(b)(i)	∞ Increase the length of the key // make key 12-bit, etc.	1
1(b)(ii)	∞ Cypher text	1

Question	Answer	Marks
1(b)(iii)	Six from: ∞ The system could use <u>odd</u> or <u>even</u> parity ∞ A parity bit is added ∞ The data is checked to see if it has incorrect/correct parity // by example ∞ If parity is correct no error is found ∞ An acknowledgement is sent that data is received correctly ∞ The next packet of data is transmitted ∞ If incorrect parity is found an error has occurred ∞ A signal is sent back to request the data is resent ∞ The data is resent until data is received correctly/timeout occurs	6

Question	Answer	Marks
3	One mark for each correct term in the correct order ∞ Serial ∞ Parallel ∞ Serial ∞ Simplex ∞ Parallel	5

Question	Answer	Marks																																				
5	One mark for each correct parity bit <table><tr><td></td><td>Parity bit</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Register A</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>Register B</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Register C</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>		Parity bit								Register A	0	0	1	0	0	0	1	1	Register B	0	0	0	0	0	1	1	1	Register C	0	0	0	0	0	0	1	1	3
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Question	Answer	Mark																																
2(a)	1 mark per each correct row: <table><thead><tr><th rowspan="2">Description</th><th colspan="2">Method</th><th colspan="3">Type</th></tr><tr><th>Serial (✓)</th><th>Parallel (✓)</th><th>Simplex (✓)</th><th>Half-duplex (✓)</th><th>Duplex (✓)</th></tr></thead><tbody><tr><td>Data is sent down a single wire in a single direction only.</td><td>✓</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>Data is sent down multiple wires in both directions, at the same time.</td><td></td><td>✓</td><td></td><td></td><td>✓</td></tr><tr><td>Data is sent down a single wire in both directions, but never at the same time.</td><td>✓</td><td></td><td></td><td>✓</td><td></td></tr></tbody></table>	Description	Method		Type			Serial (✓)	Parallel (✓)	Simplex (✓)	Half-duplex (✓)	Duplex (✓)	Data is sent down a single wire in a single direction only.	✓		✓			Data is sent down multiple wires in both directions, at the same time.		✓			✓	Data is sent down a single wire in both directions, but never at the same time.	✓			✓		3			
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2(b)(i)	1 mark for each correct parity bit: <table><thead><tr><th colspan="7">Binary Value</th><th>Parity Bit</th></tr></thead><tbody><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr></tbody></table>	Binary Value							Parity Bit	1	1	0	0	1	1	1	1	1	0	1	0	1	0	1	0	0	1	1	0	1	0	0	1	3
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2(b)(ii)	Any one from: • Transposition error // bits are interchanged • Bits still add up to even number • Even number of errors has occurred	1																																

Question	Answer	Mark
2(c)(i)	Any two from: • Scrambles data • ... making it meaningless/unintelligible • Uses an algorithm / key • Data / plain text is changed to cipher text	2
2(c)(ii)	Any one from: • Increase the length of the key // use more than 128 bits • Uses a more complex encryption algorithm	1

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Question	Answer	Marks
3(a)	Any four from: – Encryption key is used – Encryption algorithm is used – Encryption key / algorithm is applied to plain text – ... to convert it into cypher text – Same key is used to encrypt and decrypt the text	4

Question	Answer	Marks
8(a)	Any three from: – It is a universal standard – It can't be inserted the wrong way around – Supports different transmission speeds – Automatically detects if correct driver installed	3

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Question	Answer	Marks
9	– 1 – 0 – 1 – 1	4

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Question	Answer	Marks
6(a)(i)	– Uses multiple wires – Sends multiple bits of data at a time	2
6(a)(ii)	– Faster transmission speed	1

Question	Answer	Marks
9(a)	– 1 – 0 – 0 – 0	4
9(b)	Two from: – Checksum – Automatic repeat request // ARQ	2
9(c)	Any four from: – Data is input with check digit – A calculation is performed on the (inputted) data // by example – The calculated digit is compared to a stored value – If it matches, the data entered is correct – If it does not match, the data entered is incorrect	4

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Question	Answer	Marks															
2(a)	One mark for each correct row: <table border="1"> <thead> <tr> <th>8-bit binary value</th><th>Even (✓)</th><th>Odd (✓)</th></tr> </thead> <tbody> <tr> <td>11111111</td><td>✓</td><td></td></tr> <tr> <td>01100110</td><td>✓</td><td></td></tr> <tr> <td>01111011</td><td>✓</td><td></td></tr> <tr> <td>10000000</td><td></td><td>✓</td></tr> </tbody> </table>	8-bit binary value	Even (✓)	Odd (✓)	11111111	✓		01100110	✓		01111011	✓		10000000		✓	4
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01111011	✓																
10000000		✓															
2(b)	Any five from: – A value is calculated from the data – The value is calculated using an algorithm // by example – The value is appended to the data to be transmitted – Value is recalculated after transmission – Values are compared – If the values match the data is correct // if the values do not match the data is incorrect	5															

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Question	Answer	Marks
5(a)	One mark for correct tick, two marks for description – Serial – Bits sent one at a time – Single wire If parallel given, no mark for parallel, but follow through for correct description of parallel: – Multiple bits sent at a time – Multiple wires	3
5(b)	Any three from: – Universal connection // industry standard – Can only be inserted one way – Backward compatible – Auto configures // automatically recognised devices – Can power devices – Fast data transfer speed – Inexpensive to purchase/manufacture	3

Question	Answer	Marks
6	– Key // Algorithm – Algorithm // Key (must be opposite of first one) – Plain – Cypher – Key // Algorithm	5

Question	Answer	Marks															
7(a)	One mark for each correct row: <table border="1"> <thead> <tr> <th>8-bit binary value</th><th>Even (✓)</th><th>Odd (✓)</th></tr> </thead> <tbody> <tr> <td>10000001</td><td>✓</td><td></td></tr> <tr> <td>10000010</td><td>✓</td><td></td></tr> <tr> <td>00101001</td><td></td><td>✓</td></tr> <tr> <td>00101000</td><td>✓</td><td></td></tr> </tbody> </table>	8-bit binary value	Even (✓)	Odd (✓)	10000001	✓		10000010	✓		00101001		✓	00101000	✓		4
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10000010	✓																
00101001		✓															
00101000	✓																
7(b)	Any one from: – Transposition error – When bits still add up to odd/even number – Even number of incorrect bits	1															

Question	Answer	Marks
7(c)	Any one from: – ARQ – Checksum	1

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1(d)(i)	Any three from: – More accurate/reliable/efficient over long distances – Less chance of interference / cross talk – ...that will skew / distort the data // less likely to get errors – Data will arrive in order – Serial is cheaper to purchase/install/maintain	3
1(d)(ii)	– Transmission in both directions ... – ...not at the same time // asynchronous	2
1(d)(iii)	Any three from: – Calculates a value from the bits/data (to be transferred) // by example/description – Value is appended to the bits/data – Value is transferred with the bits/data – Receiver recalculates the checksum – If both values are different error is detected // if both values are the same the transmission is successful	3

2(c)(i)	Any two from: – It cannot be inserted incorrectly – Supports different transmission speeds – High speed transmission – Automatically detected (not connected) // automatically downloads drivers – Powers the device (for data transfer) – Backward compatible	2
2(c)(ii)	– Serial	1

Question	Answer	Marks
8(a)	– Odd – Odd – Even – Even	4
8(b)	Any one from: – there is a transposition of bits – it does not check the order of the bits (just the sum of 1s/0s) – even number of bits change – incorrect bits still add up to correct parity	1
8(c)(i)	Four from: – Multiple bits are sent at the same time – Uses multiple wires – Data is sent in both directions ... – ... but only one direction at a time	4
8(c)(ii)	Any two from: – Bits may arrive skewed – More expensive to setup/manufacture/purchase cable – Limited distance – More prone to interference/error	2

Question	Answer	Marks
6(a)	– Odd – Even – Even – Odd	4
6(b)	Any one from: – There is a transposition of bits – Bits still add up to correct parity	1
6(c)(i)	– Data is sent one bit at a time – Data is sent using a single wire – Data is sent in both direction ... – ... at the same time	4
6(c)(ii)	Any one from: – Data transmission can be slower (than parallel) – Additional data may need to be sent	1

NOVEMBER 2021 (1)

Question	Answer	Marks																								
3(a)	One mark per each correct row. <table><tr><th>Statement</th><th>Checksum (✓)</th><th>Check digit (✓)</th><th>Parity check (✓)</th></tr><tr><td>uses an additional bit to create an odd or even number of 1s</td><td></td><td></td><td>✓</td></tr><tr><td>checks for errors on data entry</td><td></td><td>✓</td><td></td></tr><tr><td>compares two calculated values to see if an error has occurred</td><td>✓</td><td>✓</td><td></td></tr><tr><td>will not detect transposition errors</td><td></td><td></td><td>✓</td></tr><tr><td>sends additional values when data is transmitted from one computer to another</td><td>✓</td><td></td><td>(✓)</td></tr></table>	Statement	Checksum (✓)	Check digit (✓)	Parity check (✓)	uses an additional bit to create an odd or even number of 1s			✓	checks for errors on data entry		✓		compares two calculated values to see if an error has occurred	✓	✓		will not detect transposition errors			✓	sends additional values when data is transmitted from one computer to another	✓		(✓)	5
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3(b)	– ARQ	1																								

NOVEMBER 2021 (2)

Question	Answer	Marks																																			
3(a)	One mark per each correct row. <table><thead><tr><th>Statement</th><th>Serial simplex (✓)</th><th>Parallel simplex (✓)</th><th>Parallel half-duplex (✓)</th><th>Serial duplex (✓)</th></tr></thead><tbody><tr><td>bits are transmitted along a single wire</td><td>✓</td><td></td><td></td><td>✓</td></tr><tr><td>data is transmitted in both directions</td><td></td><td></td><td>✓</td><td>✓</td></tr><tr><td>it is only suitable for distances less than 5 metres</td><td></td><td>✓</td><td>✓</td><td></td></tr><tr><td>Bits from the same byte are transmitted one after the other</td><td>✓</td><td></td><td></td><td>✓</td></tr><tr><td>data may not arrive in the correct sequence</td><td></td><td>✓</td><td>✓</td><td></td></tr><tr><td>data is transmitted in both directions, but only one direction at a time</td><td></td><td></td><td>✓</td><td></td></tr></tbody></table>	Statement	Serial simplex (✓)	Parallel simplex (✓)	Parallel half-duplex (✓)	Serial duplex (✓)	bits are transmitted along a single wire	✓			✓	data is transmitted in both directions			✓	✓	it is only suitable for distances less than 5 metres		✓	✓		Bits from the same byte are transmitted one after the other	✓			✓	data may not arrive in the correct sequence		✓	✓		data is transmitted in both directions, but only one direction at a time			✓		6
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data is transmitted in both directions, but only one direction at a time			✓																																		
3(b)	Any three from: – Can charge/power the mobile device (at the same time) – (Uses serial transmission so) data less likely to be skewed / corrupted – Universal / industry standard / connection – Cable can only be plugged in one way // Cannot be inserted incorrectly – Fast transmission speed – Backward compatible – Supports different transmission speeds – Automatically detects device // Automatically downloads drivers	3																																			

Question	Answer	Marks																								
3(a)	One mark per each row <table><tr><th>Statement</th><th>ARQ (✓)</th><th>Check digit (✓)</th><th>Checksum (✓)</th></tr><tr><td>checks for errors on data entry</td><td></td><td>✓</td><td></td></tr><tr><td>uses a process of acknowledgement and timeout</td><td>✓</td><td></td><td></td></tr><tr><td>compares two calculated values to see if an error has occurred</td><td></td><td>✓</td><td>✓</td></tr><tr><td>may resend data until it is confirmed as received</td><td>✓</td><td></td><td></td></tr><tr><td>checks for errors in data after transmission from a computer to another</td><td></td><td></td><td>✓</td></tr></table>	Statement	ARQ (✓)	Check digit (✓)	Checksum (✓)	checks for errors on data entry		✓		uses a process of acknowledgement and timeout	✓			compares two calculated values to see if an error has occurred		✓	✓	may resend data until it is confirmed as received	✓			checks for errors in data after transmission from a computer to another			✓	5
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3(b)	– Parity check	1																								

Question	Answer	Marks
4(c)(i)	Four from: – Bits sent one at a time – ... down a single wire – Data sent in both directions ... – ... but only one direction at a time	4
4(c)(ii)	Any two from: – Simplex only sends data in one direction – ... so, printer may not be able to tell computer an error has occurred, and computer may not be able to send printer the document to be printed NOTE: Award any valid contextual answer for MP2	2

Question	Answer	Marks
9	– Even – Even – Odd – Even	4

Question	Answer	Marks
2(c)(iii)	Any five from: • Parity can be set to odd or even • Sender and receiver agree on parity to use • Data/email/image is split into bytes // blocks of 7 bytes • Sender counts the number of 1s/0s in each group/byte • Each group/byte is assigned a parity bit to match the parity/odd/even • Receiving device/server recounts the number of 1s/0s in each group/byte • ... and compares to parity used/odd/even and if it does not match the parity, an error is reported/identified • (in block check) the location of the error(s) can be identified/estimated at the intersection	5
2(c)(iv)	1 mark for identification Any three for corresponding description: • Checksum • Value is calculated from the data ... • ...using an algorithm // by example • Value is transmitted with the data • Value recalculated by receiver (using same algorithm) ... • ...if checksum values are different there is an error // reverse • Automatic Repeat request/reQuery (ARQ) • Uses acknowledgement and timeout • Sender starts a timer when data is transmitted • Receiver uses an error checking method to check whether the data has been received accurately • If no error detected a positive acknowledgement is returned to sender • If error detected negative acknowledgement is returned to sender • If sender gets no acknowledgement within the set time it resends the data	4

JUNE 2022(3)

Question	Answer	Marks
7(a)	• Data is sent multiple bits at a time • ... down multiple wires • Data is sent in one direction only	3
7(b)	Any two from: • Parallel is not suitable for long distances // only suitable for short distances • ... the data could become skewed • Parallel cables are not manufactured above approx. 5m	2
7(c)	Any three from (MAX 2 for ARQ): • Checksum used to detect errors (during transmission) • ... using a calculated value • ARQ checks if data is received • ... uses acknowledgement and timeout • ... requests data be sent again if (checksum) detects error / not received	3

MARCH 2023

Question	Answer	Marks																
5(a)(i)	Any two from: e.g. • Destination IP/address • Packet number • Originators IP/address • Error detection method	2																
5(a)(ii)	One mark each: • Payload • Trailer	2																
5(b)(i)	No mark for choice. Three from for justification that matches choice. Serial • Data arrives in order sent // does not need reordering • Less likely to experience interference • ... less likely to have errors • Can transmit over a longer distance (i.e. another room) • Still fast transmission... • ...sufficient for this purpose Parallel • Faster transmission speed than serial • ... faster response to requests • Very long connection not needed • ...next room is (likely) within distance for parallel • ... unlikely to error/arrive out of sequence/skew	3																
5(b)(ii)	One mark each: • Data goes in both directions... • ...at the same time	2																
5(c)(i)	One mark each: <table border="1"><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>	0	1	1	0	0	0	1	1	0	0	0	0	0	0	0	0	2
0	1	1	0	0	0	1	1											
0	0	0	0	0	0	0	0											
5(c)(ii)	One mark each: • In parity check, interchange of bits will not be detected // Parity check cannot detect even number of changes // Parity check cannot detect error if parity stays correct ... • ...the (possible) position of all changes will be highlighted // will identify the horizontal and vertical position of all differences/changes	2																
5(c)(iii)	One mark each: • Bit 6 • Byte 4	2																

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2(c)(i)	Any two from: e.g. • Destination/receivers (IP) address • Packet number • Originator's/senders (IP) address	2
2(c)(ii)	Any five from: • Data is broken/split/divided into packets • Each packet (could) take a different route • A router controls the route/path a packet takes • ... selecting the shortest/fastest available route/path • Packets may arrive out of order • Once the last packet has arrived, packets are reordered • If a packet is missing/corrupted, it is requested again	5

JUNE 2023 (2)

Question	Answer	Marks
4(a)(i)	Two from: • Data is sent one bit at a time • A single wire is used	2
4(a)(ii)	Any two from: • Data won't be skewed • Less chance of interference/crosstalk/corruption/error • Transmission speed is adequate	2
4(a)(iii)	• The data may be transmitted quicker	1

JUNE 2023 (3)

Question	Answer	Marks
5	The diagram demonstrates (one mark for each): • Packets sent through several routers • ... taking different routes from device A to device B • Packets arrive out of order • Packets being reordered when all arrived at device B	4

NOVEMBER 2023

Question	Answer	Marks
4(a)	One mark for each correct transmission method: – Serial half-duplex – Serial full-duplex	2
4(b)	Any four from: – The number of 1 s/0 s are counted – A parity bit is added to each byte/7 bits before transmission – ... to make the sum of the bits/1 or 0 s in each byte odd – After transmission, if the number is odd no error is detected – After transmission, if the number is even an error is detected	4

Question	Answer	Marks
4(c)	– Echo (check)	1

Question	Answer	Marks
7	One mark for each correct term, in the correct order: – header – destination address – routers – last	4

Question	Answer	Marks										
6	One mark for each correct data transmission method: <table><tr><th>Data transmission method</th><th>Description</th></tr><tr><td>serial simplex</td><td>Data is transmitted down a single wire, one bit at a time, in one direction only.</td></tr><tr><td>parallel half-duplex</td><td>Data is transmitted down multiple wires, multiple bits at a time, in both directions, but only one direction at a time.</td></tr><tr><td>serial full-duplex</td><td>Data is transmitted down a single wire, one bit at a time, in both directions at the same time.</td></tr><tr><td>parallel simplex</td><td>Data is transmitted down multiple wires, multiple bits at a time, in one direction only.</td></tr></table>	Data transmission method	Description	serial simplex	Data is transmitted down a single wire, one bit at a time, in one direction only.	parallel half-duplex	Data is transmitted down multiple wires, multiple bits at a time, in both directions, but only one direction at a time.	serial full-duplex	Data is transmitted down a single wire, one bit at a time, in both directions at the same time.	parallel simplex	Data is transmitted down multiple wires, multiple bits at a time, in one direction only.	4
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NOVEMBER 2023 (3)

Question	Answer	Marks
5(a)	– Interference // crosstalk	1
5(b)	– C	1
5(c)	Any five from: – Timer is started when sending device transmits a data packet to receiver – Receiving device checks the data packet for errors – Once the receiving device knows the packet is error free it sends an acknowledgement back to the sending device ... – ... and the next packet is sent – If the sending device does not receive an acknowledgement before the timer ends ... – ... a timeout occurs – ... the data packet is resent ... – ... until acknowledgement received // until max number of attempts reached	5